

**AMENDMENT NO. 2 FEBRUARY 2007
TO
IS 6921 : 1973 METHODS OF SAMPLING AND
TEST FOR LAC AND LAC PRODUCTS**

(Page 56, clause 27) — Insert the following after clause 27:

**28 UV AND VISIBLE SPECTRAL ANALYSIS FOR ASSERTION OF
QUALITY OF SEEDLAC, SHELLAC AND BLEACHED LAC**

28.1 General

Shellac (lac resin) contains polyhydroxyanthraquinone derivative pigments, of erythrolaccin group. The constituent acids of the resin also contain double bonds. These lead to characteristic absorption maxima in visible and UV range.

During the bleaching process, the chromophoric groups of the colouring pigments are destroyed, resulting in absence of absorption maxima in the visible region.

28.2 Outline of the Method

The method involves running of wavelength scan (absorbance vs. wavelength) of alcoholic solution of shellac or bleached lac for detection of presence/absence of certain absorption maxima in the UV and visible range.

28.3 Apparatus

- a) Any spectrophotometer capable of measuring absorbance in the range 190/200-700 nm. Equipment with wavelength scan facility with printer would be preferred.
- b) Volumetric flask with stopper 100 ml, 10 ml (3 Nos.)
- c) Pipettes (graduated) — 2 and 1 ml
- d) A weighing balance (accuracy ± 0.1 mg), Whatman filter, glass funnel, etc.
- e) Quartz cuvettes, 1 cm path length.

28.4 Reagent

Ethyl alcohol (absolute)

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28.5 Procedure

28.5.1 Weigh accurately 1 g of finely powdered shellac/bleached lac and transfer to 100-ml volumetric flask. Add about 60 - 70 ml of ethyl alcohol; place the stopper and shake vigorously till the material is dissolved. Add solvent up the mark of the flask. Filter the solution using Whatman No. 1 filter paper preferably under saturated vapour pressure conditions. Discard the first 15 ml of the clear filtrate. The concentration is 10^{-2} g/ml.

28.5.2 *Solution for Absorption Spectrum in the Visible Range*

Transfer 1.5 ml of the filtrate (**28.5.1**) by means of a 2-ml pipette to a 10-ml volumetric flask. Add ethyl alcohol up to the mark. The concentration becomes 1.5×10^{-3} g/ml.

28.5.3 *Solution for Absorption Spectrum in the UV Range*

Transfer 1 ml of the filtrate (**28.5.1**) using a 1 ml pipette to a 10-ml volumetric flask. Add ethyl alcohol to it up to the mark on the flask. The concentration of the solution is now 10^{-3} g/ml. Now again transfer 1 ml of the diluted solution to another 10 ml flask using a 10 ml pipette and make up the volume up to the mark using ethyl alcohol. The concentration is now 10^{-4} g/ml.

28.5.4 *Measurement of Absorption Spectrum in the Visible Range*

Switch on the spectrophotometer. Allow to warm up. Set the wavelength at 700 nm. Match the cuvettes with the alcohol to be used for the preparation of the solution. Transfer required quantity of diluted solution outlined under **28.5.2** to one of the cuvettes. Run a wavelength scan between 400-700 nm for absorbance. Study the spectrum.

Shellac: Absorption peak at 425 - 430 nm.

Bleached lac: The above peak will be absent.

28.5.5 *Measurement of Absorption Spectrum in the UV Range*

Set the wavelength at 400 nm. Match the cuvettes with the alcohol to be used for the preparation of the solution. Transfer required quantity of diluted solution outlined under **28.5.3** to one of the cuvettes. Run a wavelength scan between 190/200 – 400 nm for absorbance. Study the spectrum.

If the absorption values exceed the upper limit, a solution diluted further may be used for the run.

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Shellac: Absorption peak at 225, 285 nm and shoulder at 215 nm.

Bleached lac: Peaks at 215, 225, 290-300 nm; small peaks at 255, 265 nm.

28.6 Report

Presence/absence of peaks and shoulders at the wavelengths specified above are to be mentioned.

29 DSC ANALYSIS FOR EVALUATING THE QUALITY

29.1 General

The softening (glass transition) and melting temperatures of different forms of lac resin can be determined from their melting profiles (a plot of heat flow against temperature) employing a Differential Scanning Calorimeter. An analysis of the profiles by appropriate software provides accurate and reliable information free from subjectivity. The DSC analysis also provides information on occurrence of resin and wax and purity. The presence of the wax can be determined in shellac, dewaxed lac, bleached lac, etc.

29.2 Apparatus

A computer controlled Differential Scanning Calorimeter with provision for different rates of heating and cooling (a minimum of 2°C/min with software for peak position, selection of ranges of dependent and independent variables, determination of glass transition, melting temperatures, partial area, percentage of solid present, purity determination, etc.

The temperature range required for the analysis is 10-120°C.

- a) Printer, printing paper for the output from DSC analysis*
- b) Aluminium pans with cover for DSC analysis with crimper⁴¹
- c) A weighing balance with accuracy of ± 0.01 mg.

29.3 Procedure

29.3.1 Weigh finely powdered 5 - 6mg of lac resin sample, accurately, in the aluminium pan, put the cover and crimp properly.

29.3.2 Switch on the printer, DSC (precalibrated) and controller. Start purging the sample holder with Ar or N₂. Start the compressor/circulator.

*The equipment/material would normally be provided with DSC.

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29.3.3 Place the sample in the crimped pan in the sample holder of DSC.

29.3.4 Start the software, select the appropriate mem for a temperature profile run. Enter the sample, file name, etc. Enter start and end temperatures, run rate (20°C/min), etc. For different forms of lac resin, the starting temperature should not be more than 20-25°C, as far as possible.

29.3.5 After attaining the steady state, start run, save the profile after completion of the run.

29.3.6 Enter into analysis software, select the appropriate portion of the profile, preferably containing the initial and final plateau regions.

29.3.7 *Glass Transition Temperature*

Open the Tg menu in the software. Position the cursors in the regions. Enter the Tg value. Repeat a few more times by selecting different regions. Selection of Tg is to be made a set of nearly similar values. Care is to be exercised that Tg value is less than the onset temperature if onset is also available in the same menu.

29.3.8 *Melting Temperature*

Position of the melting peak will be shown on the screen, which can be taken as melting temperature.

29.3.9 *Purity*

Open the Purity menu. Enter the values of weight of the sample and molecular weight. Select proper regions. Select purity calculation. Repeat the operation for different regions. Select the purity value from a set of near similar values.

29.4 Examination of Melting Profile

29.4.1 *Seedlac, Shellac, Regular Bleached Lac*

Two melting peaks are obtained, at 55-57°C and 72-74°C, due to resin and wax, respectively. The positions may somewhat vary among samples. For bleached lac the resin peak may be found a few degrees higher compared to seedlac and shellac. Since lac is of natural origin, the positions of melting peaks of resin and wax have been found to vary by a few degrees as mentioned above. Reporting may be made about the actual temperatures of peaks observed for resin and wax.

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29.4.2 *Dewaxed Lac, Dewaxed Bleached Lac*

Single melting peak would be found at 55-57°C for dewaxed lac. There will be no peak corresponding to wax. For dewaxed bleached lac the resin peak may be found a few degrees higher compared to seedlac and shellac. No peak for wax (at around 72-74°C) will be observed for dewaxed bleached lac also.

29.4.3 *Kiri*

Two melting peaks (at 55-57°C and 72-74°C) would be obtained. The height of the resin peak would be lower compared to wax peak. The position of the peaks may be a few degrees higher than seedlac or shellac.

29.4.4 *Old Seedlac/Shellac*

Polymerized lac does not give any melting peak for the resin. The peak corresponding wax would be found in 72-74°C.

29.5 Report

The report should include glass-transition (softening) and melting temperatures (in °C); position of peaks (in °C) and number.

NOTE — For seedlac/shellac, well defined peaks for resin and wax may not be obtained at the first instance in some cases. In order to find well defined peaks, the particular seedlac/shellac sample is to be heated up to 120°C, then a cooling run is to be made preferably at a very slow rate (1 – 2°/min) from 120°C to 20/25°C, then the melting run is to be performed as per the method laid down in procedure in 29.3.4 onwards. The profiles will definitely provide better information about the melting peaks.

(CHD 23)